

ECD–QV Pipeline

Gate Parameter Optimizer

`ECDParameterFinder` · batched Adam
N Haar targets · ideal ECD model

$\{\beta_j, \theta_j\}$ per ECD layer

Pulse Builder

`ECDPulseBuilder` · solve $\alpha(t)$ ODE
virtual-Z correction · cavity phase

$\varepsilon(t), \omega(t)$ waveforms

Displaced Frame Simulation

`DisplacedFrameSimulator` · QuTiP / dynamiqs
T1/T2 noise sweep · HOG & XEB metrics